

SmartDialer Demonstration & Operations Guide

Interactive Control Center Walkthrough, Chaos Injections & Reviewer Evaluation

1. Execution Quick-Start

SmartDialer can be executed via the Streamlit Operations Dashboard or via standalone CLI scripts.

Target Component	Command Line Execution	Description & Output
Streamlit Dashboard	<code>python -m streamlit run frontend/app.py</code>	Launches the interactive dark-themed operations dashboard on <code>http://localhost:8501</code> .
Full Test Suite	<code>python -m pytest tests/ -q</code>	Executes all 202 automated unit and integration tests (~25–35 seconds).
High-Concurrency Load Test	<code>python scripts/load_test.py</code>	Runs 100, 1k, and 10k agent reservation throughput benchmarks (up to 44k ops/sec).
Live CLI Demo	<code>python scripts/demo.py</code>	Fast 5-second end-to-end terminal demo printing cycle tables and statistics.
Scenario Simulation	<code>python scripts/simulation.py</code>	Runs Progressive vs Predictive vs Chaotic Provider B side-by-side comparison.

2. Streamlit Control Center Navigation Tour

Page / View	Key Capabilities & Interactive Controls
1. Dashboard	6 KPI cards, live agent donut chart, pacing time series, recent safety decisions & live calls.
2. Borrowers & Queue	Priority-sorted queue (HIGH/MED/LOW), attempt counts, and interactive 'Enrol Borrower' form.
3. Live Calls	Filter by provider/state, inspect version counters, and 'Initiate Single Direct Call' dispatcher.
4. Predictive Engine	4-stage pipeline math trace (Conditions → Prediction → Safety → Approved) + What-If Sandbox.
5. Safety Controller	Audit log with reason codes, decision breakdown chart (APPROVE, REDUCE, REJECT).
6. Providers	Carrier status (Provider A clean vs Provider B chaos), Circuit Breaker telemetry, drop metrics.
7. Chaos & Failures	Live fault buttons: Trip Circuit Breaker, Drop N Agents, Inject Crashed Worker, Restore System.
8. Simulation	Benchmark Scenarios A, B, C, D execution with full Plotly performance analytics.

3. Fault-Tolerance Demonstration Scenarios

Scenario 1: Carrier Outage Injection

Trigger 'Trip Circuit Breaker' on Page 7 → Observe breaker state switch to OPEN → Trigger a call dispatch → Verify Safety Controller REJECT with reason '*Circuit breaker is OPEN*'.

Scenario 2: Shift End / Agent Availability Drop

Trigger 'Drop 10 Available Agents' → Observe available pool shrink → Trigger Pacing Request → Verify Safety Controller clamps approved calls to remaining live capacity.

Scenario 3: Worker Server Crash

Trigger 'Simulate Worker Crash' → Injects call with expired lease → Click 'Run Reconciler' → Verify Reconciler reclaims

orphaned agent back to AVAILABLE while protecting connected calls.

Scenario 4: Provider B Chaos Webhooks

Switch to Provider B → Observe duplicates & out-of-order events ingested → Verify Event Processor drops duplicates in $O(1)$ with 0 state corruption.